

BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

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Abstract

Recovering prior client-specific advice from a growing legal consultation archive is difficult under same-domain interference. Dense retrieval often ranks the client’s question highly while missing the lawyer’s answer—an episode-incompleteness failure—and product-quantized indexes can further distort turn scores at moderate archive size, fragmenting consultation episodes across unrelated top- k slots in a vector database. BIMS-LEGAL is a dual-store Brain-Inspired Memory System for Chinese legal consultation logs. Its systemic contribution is a complementary *episodic store* (timestamped turns with role and session identifier *sid*) paired with a slower *BIRCH semantic store* of thematic clusters, separating fast episode recovery from cross-session semantic integration. Algorithmically, Soft O2 (session soft binding) and Soft O2-C with gated Hybrid cluster binding address episode incompleteness under IVFPQ score collapse and context fragmentation: siblings inherit attenuated scores ($\beta \cdot s$) rather than hard score copy, completing question-hit / answer-miss episodes while preserving rank competition. O1 (exact FlatIP indexing) and O3 (deferred bulk-load consolidation) are implementation and scale design operators that enable rigorous large-scale shared-store experiments; they are not standalone algorithmic contributions. We evaluate on CAIL2024 multi-turn dialogues and on LegalEp episode corpora rebuilt

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from DISC-Law-SFT and Lawyer-LLaMA, with exact replay as a diagnostic channel and paraphrase, follow-up, and advice-recall as primary channels. Controls include FlatIP, hard parent hydration, BM25, dense-sparse RRF, and cross-encoder reranking, with McNemar tests and bootstrap intervals. On the $M=400$ shared store, O1+Soft O2 raises Answer Hit from 0.540/0.397 to 0.780/0.680 on DISC/Lawyer. On CAIL multi-turn Soft O2 lifts Answer Hit by +0.22 to +0.52 over FlatIP with collapsing shuffled-*sid* controls. On same-session LegalMem/LegalEp stores Soft O2 dominates ungated Soft O2-C; under a fair Mix reconstruction of the same stores, gated Hybrid Soft O2+Soft O2-C improves over Soft O2 on LegalEp-Lawyer (AH 0.534 vs 0.496, mid- $p=0.010$).

Keywords: legal information retrieval, consultation memory, soft episode binding, dual-store memory, cluster binding, Chinese legal corpora

1. Introduction

A junior associate may ask a firm assistant: “Last month you advised this client on terminating a fixed-term contract—what remedy did we recommend?” The system returns several near-identical “contract termination” questions, ranks the stored user utterance highly, and never surfaces the partner’s written advice. The subsequent reply may sound fluent yet be grounded in the wrong episode—or in none. In consultation archives, questions and answers share specialized terminology; neighbouring matters are topically adjacent yet legally distinct; and the professionally useful evidence unit is the full consultation *episode*.

Legal informatics has long studied statutory interpretation, case retrieval, and advisory systems Martinez-Gil (2023); Ashley (2017). Recent legal large language models (LLMs) improve statutory and fact-pattern answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical legal information retrieval (IR) mainly targets *external* authority—statutes, cases, or community answers—as documents Askari et al. (2024); Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented dialogue systems recover evidence from evolving conversation logs Wu et al. (2024); Maharana et al. (2024); Packer et al. (2023). The gap addressed here is *internal consultation memory*: retrieving prior client-specific advice from a growing same-domain archive under lexical interference.

BIMS-LEGAL is a dual-store Brain-Inspired Memory System for legal

consultation archives, designed and evaluated in this study. The *systemic contribution* is a dual-store architecture that separates a fast episodic pathway (timestamped turn embeddings with speaker role and session identifier *sid*) from a slow semantic pathway based on Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH) with deferred consolidation Zhang et al. (1996). Complementary Learning Systems (CLS) theory McClelland et al. (1995); Kumaran et al. (2016) supplies an architectural analogy for this dual-pathway design, not an empirical claim about biological memory. Under same-domain interference, dense turn retrieval in a vector database often returns a question turn while its answer sibling remains outside top- k —a system-level episodic memory failure we term *episode incompleteness*, distinct from a full session miss and exacerbated by IVFPQ score collapse and context fragmentation at moderate archive sizes.

The *algorithmic contributions* are soft binding operators that complete episodes without hard score copy:

- **Soft O2 (session soft binding).** When a turn is retrieved with score s , siblings sharing *sid* inherit $\max(s_{\ell}, \beta \cdot s)$, surfacing answer turns under question-hit / answer-miss while preserving rank competition among candidates.
- **Soft O2-C and gated Hybrid (cluster soft binding).** The same soft-inheritance rule applies within BIRCH clusters when gold evidence is not co-session with the query; Hybrid triggers cluster binding only on direct dense hits to avoid displacing same-session siblings.

Two **implementation and scale design operators** support rigorous large-scale shared-store experiments but are not standalone algorithmic contributions:

- **O1 (exact FlatIP indexing).** At hundreds-to-thousands of turns, IVFPQ product quantization is under-trained and collapses Answer Hit; FlatIP restores faithful turn scores so Soft O2 inherits uncorrupted sibling signals.
- **O3 (bulk-load consolidation).** Deferred BIRCH and disk writes make archive construction tractable without changing retrieval semantics.

We ask four research questions:

- 57 • **RQ1.** Under same-domain interference, does episode incompleteness
58 (question hit / answer miss) dominate full session miss?
- 59 • **RQ2.** Do Soft O2 and Soft O2-C address episode incompleteness under
60 measured index distortion, and what incremental gains does Soft O2
61 provide over FlatIP and hard parent hydration?
- 62 • **RQ3.** On the *same* LegalEp /LegalMem stores, when is the slow
63 (BIRCH) pathway effective? Does Soft O2-C help under standard same-
64 session gold, and under a fair Mix of same-/cross-session gold?
- 65 • **RQ4.** Across CAIL multi-turn and LegalEp paraphrase /advice-recall
66 channels, how large and session-tied are Soft O2 gains, and how do
67 Soft O2 and Soft O2-C /Hybrid complement each other?

68 *Contributions..*

- 69 1. A dual-store memory architecture (episodic + BIRCH semantic) for
70 Chinese legal consultation archives, separating fast episode recovery
71 from cross-session semantic integration under same-domain interfer-
72 ence.
- 73 2. Soft O2 (session soft binding) as a system-level solution to episodic
74 memory incompleteness under IVFPQ collapse and context fragmen-
75 tation in vector databases, with Soft O2-C and gated Hybrid cluster
76 binding for cross-session episode recovery.
- 77 3. Same-store and fair Mix validation showing when each pathway applies:
78 Soft O2 dominates when gold shares *sid*; gated Hybrid Soft O2+Soft O2-
79 C improves under cross-session Mix reconstructions (LegalEp-Lawyer
80 AH 0.534 vs 0.496, mid- $p=0.010$).
- 81 4. Multi-channel evidence on CAIL and LegalEp with BM25, RRF, and
82 cross-encoder controls, McNemar tests, scale curves to $M=6400$, and a
83 dual-protocol end-to-end QA audit linking complete episode retrieval
84 to reduced hallucination.
- 85 5. O1 and O3 as implementation and scale design operators enabling the
86 shared-store ablations ($M=400$, $S=300$) and archive construction at
87 scale, reported separately from the algorithmic claims.

88 Section 2 reviews related work. Section 3 presents BIMS-LEGAL, Soft O2,
89 Soft O2-C, and implementation operators O1/O3. Section 4 details corpora
90 and protocols. Section 5 reports results. Section 6 discusses implications and
91 limitations. Section 7 concludes.

2. Related Work

2.1. Legal IR and long-horizon dialogue memory

Legal QA and retrieval systems typically ground generation in statutes, cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware similar-case ranking further show how interaction context and structured legal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024). Those pipelines optimize access to external authority. Consultation memory retrieval targets a different evidence source: *prior consultation episodes* inside an agent’s own archive. Confusing two clients’ histories can produce fluent yet professionally inappropriate responses even when statute retrieval is correct. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM improve statutory and advisory generation Yue et al. (2023); Huang et al. (2023); Yue et al. (2024), but they still need a reliable memory index when prior advice must be recovered from a growing same-domain log.

Legal consultation archives add a further difficulty: high lexical overlap among topically adjacent matters, with the professionally useful evidence unit being the full question–answer episode. This setting motivates *dual-store* memory architectures that keep fast episodic turn retrieval separate from slower semantic clustering, and session-aware soft binding operators that recover answer turns under same-domain interference without collapsing rank competition among unrelated candidates.

2.2. Dual-store memory and episodic retrieval

Dual-store designs distinguish fast episodic encoding from slower semantic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving et al. (1972); Atkinson and Shiffrin (1971). Memory architectures for dialogue agents extend context via hierarchical paging, summary banks, graphs, and neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Rasmussen et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang et al. (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved units as flat passages rather than *multi-turn consultation episodes* whose answer turn may rank below an unrelated question turn sharing legal vocabulary. LongMemEval and LoCoMo evaluate long-horizon recall and QA over personal conversation logs Wu et al. (2024); Maharana et al. (2024), while broader memory-agent surveys emphasize persistence beyond the context window Wu et al. (2025); Tan et al. (2025). Recent conversational

128 RAG work stresses reuse of historical search evidence across turns Mo et al.
 129 (2026). BIMS-LEGAL positions its dual-store contribution at this intersec-
 130 tion: episodic turn embeddings with *sid* for same-session recovery (Soft O2),
 131 and BIRCH centroids with Soft O2-C for cross-session recovery when query
 132 and gold evidence occupy different sessions but share thematic structure
 133 Zhang et al. (1996). Soft O2 directly addresses the episodic retrieval chal-
 134 lenge that parent-document hydration and session-max aggregation handle
 135 via hard score copy: under dense same-domain interference, a question turn
 136 can dominate top- k while its answer sibling remains absent, producing fluent
 137 yet ungrounded downstream generation unless siblings enter under attenu-
 138 ated inherited scores.

139 2.3. Session-aware retrieval

140 Parent-document hydration, session aggregation, and joint session index-
 141 ing are standard session-aware mechanisms for attaching sibling context to
 142 a dense hit. Soft episode binding (Soft O2) inherits $\beta \cdot s$ for siblings and
 143 keeps ranking competition among candidates. Lexical BM25 Robertson and
 144 Zaragoza (2009), dense passage retrieval Karpukhin et al. (2020), dense-
 145 sparse Reciprocal Rank Fusion Cormack et al. (2009), and cross-encoder
 146 reranking Nogueira et al. (2019); Chen et al. (2024) provide complemen-
 147 tary controls outside session membership. Exact flat indexes and product-
 148 quantized ANN backends Johnson et al. (2019) further affect whether sibling
 149 binding is applied over faithful turn scores. Table 1 summarizes the session-
 150 structure contrasts used in the main grids.

Table 1: Session-aware mechanisms compared in this work.

Mechanism	Unit	Expansion	Type	Rank
Parent hydration	Session	Insert siblings	Hard	No
Session-max	Session	Session max	Hard	Same*
Joint episode index	Session doc	Atomic	N/A	N/A
Soft O2	Turn+ <i>sid</i>	Soft βs	Soft	Yes
Soft O2-C	Turn+cluster	Soft $\beta_c s$	Soft	Yes

151 *On the reported corpora, session-max rankings coincide with hard parent hydration
 152 under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2
 153 denotes session soft binding; Soft O2-C denotes cluster soft binding on the BIRCH path-
 154 way.

155 We reuse CLS vocabulary as an architectural analogy for BIMS-LEGAL:
 156 turn embeddings with timestamps, roles, and session identifiers as episodic
 157 elements; Soft O2 as episode binding on the episodic pathway; BIRCH cen-
 158 troids with Soft O2-C as semantic integration on the slow pathway; and
 159 shared-store distractors as interference. The dual-store split is the systemic
 160 design; Soft O2 and Soft O2-C are the algorithmic mechanisms evaluated on
 161 Chinese legal consultation archives.

162 3. Method: BIMS-LEGAL

163 3.1. Architecture overview

164 Figure 1 summarizes the pipeline. BIMS-LEGAL writes each consulta-
 165 tion turn as an episodic element and maintains a semantic store of thematic
 166 clusters. The episodic store supports fast encoding with timestamps, speaker
 167 roles, and session identifiers. The semantic store aggregates related turns into
 168 centroids through deferred BIRCH consolidation during bulk archive con-
 169 struction (O3). Same-session Soft O2 grids use the episodic pathway; Mix
 170 Soft O2-C grids additionally exercise cluster soft binding on the semantic
 171 pathway.

172 Retrieval combines episodic dense similarity with optional associative and
 173 temporal cues on the turn index. Soft O2 binds siblings that share a session
 174 identifier; Soft O2-C binds siblings that share a BIRCH cluster identifier.

175 **Definition 1** (Episodic and semantic elements). *An episodic element is*
 176 $e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, sid_i)$, *where* $\mathbf{v}_i \in \mathbb{R}^d$ *is an embedding,* t_i *a timestamp,* c_i
 177 *conversational context (including speaker role),* ϕ_i *surface features, and* sid_i
 178 *the consultation session identifier. A semantic element is* $s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$
 179 *with centroid* $\mathbf{c}_j$, *members* C_j , *label* κ_j , *and formation statistics* τ_j .

180 3.2. Implementation and scale design (O1, O3)

181 For query q with embedding $\mathbf{v}_q$, episodic base retrieval scores a memory
 182 turn m by

$$\text{score}(m) = \alpha S(\mathbf{v}_q, \mathbf{v}_m) + (1 - \alpha) R(t_m), \quad (1)$$

183 where S is cosine similarity, $R(t) = \exp(-\lambda \Delta t)$, and $\alpha=0.7$ by default. This
 184 additive semantic-recency fusion is the ranking score used throughout the
 185 legal grids; session soft inheritance (Soft O2) is a subsequent binding step
 186 and does not replace Eq. (1).

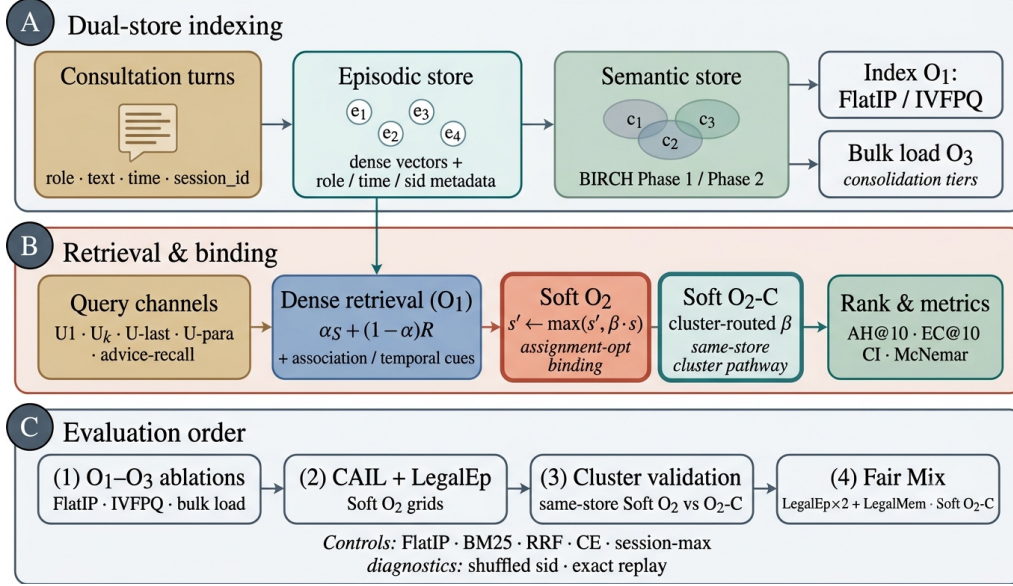


Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and BIRCH semantic clustering (O3 bulk-load consolidation). (B) Query-time dense retrieval under O1 FlatIP with Soft O2 session binding and Soft O2-C cluster binding (algorithmic core); O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, CAIL /LegalEp Soft O2 grids, same-store cluster validation, and fair Mix Soft O2-C.

187 *O1: Exact FlatIP indexing (implementation operator)..* At hundreds to a
 188 few thousand turns, FAISS IVFPQ is often under-trained. Quantization
 189 error depresses Answer Hit (Table A.8: IVFPQ AH 0.330 at $M=1600$ on
 190 LegalEp-DISC exact). O1 disables product quantization and retrieves with
 191 exact FlatIP so Soft O2 inherits faithful turn scores. O1 is an engineering
 192 prerequisite for rigorous Soft O2 evaluation, not a standalone retrieval algo-
 193 rithm.

194 3.3. Soft O2: session soft binding

195 *Motivation..* Under same-domain interference in a turn-level vector database,
 196 a retrieved question turn can score highly while its answer sibling remains
 197 outside top- k —episode incompleteness compounded by IVFPQ score collapse
 198 and context fragmentation across unrelated candidates. Soft O2 is a system-
 199 level binding step that completes episodes without hard score copy.

200 *Soft O2 rule..* When turn t is retrieved with score s_t , siblings sharing sid_t
 201 receive

$$s_{t'} \leftarrow \max(s_{t'}, \beta \cdot s_t). \quad (2)$$

202 Default $\beta=0.98$, with sensitivity over $\{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$. Fair con-
 203 trols include hard parent hydration, session-max aggregation, and shuffled
 204 *sid*. On two-turn LegalEp episodes, hard hydration and session-max coin-
 205 cide under full-sibling expansion; we report a single hard-expansion baseline.
 206 Dense joint episode indexing is an atomic-index bound with a different evi-
 207 dence unit.

208 *O3: Bulk-load consolidation (implementation operator)..* During bulk con-
 209 struction, O3 defers per-insert BIRCH updates and disk flushes to a single
 210 **finalize_bulk_load** pass. O3 does not change retrieval semantics; it makes
 211 $M \geq 400$ shared-store experiments runnable (≈ 11 min wall-clock for $M=400$;
 212 amortized ≈ 0.825 s/turn over ~ 800 turns). We correct an earlier arithmetic
 213 misstatement and do not treat O3 wall-clock as evidence of asymptotic scal-
 214 ability; quality-latency curves to $M=6400$ appear in Section 5.4.

215 3.4. Semantic consolidation and Soft O2-C

216 Semantic consolidation follows a BIRCH-style dual-phase procedure Zhang
 217 et al. (1996). Phase 1 assigns a new embedding to the nearest centroid above
 218 threshold θ_H , or creates a cluster when no centroid qualifies. Phase 2 period-
 219 ically merges high-similarity centroids above θ_M and purges near-duplicates.
 220 In BIMS-LEGAL this pathway runs under O3 deferred consolidation; the
 221 resulting cluster membership is the substrate for Soft O2-C.

222 Soft O2-C mirrors Soft O2 on the semantic store. When turn t is retrieved
 223 with score s_t , siblings sharing BIRCH cluster identifier cid_t receive

$$s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_t). \quad (3)$$

224 Default $\beta_c=0.95$, with a per-cluster sibling budget to limit large thematic
 225 clusters from flooding top- k . A gated Hybrid Soft O2+Soft O2-C injects
 226 only cross-session siblings relative to sessions already covered by dense hits,
 227 triggered on direct dense hits. Soft O2 and Soft O2-C are complementary:
 228 Soft O2 is the default binder when query and gold share *sid*; Soft O2-C is
 229 required when gold evidence is held in a different session but the same cluster.

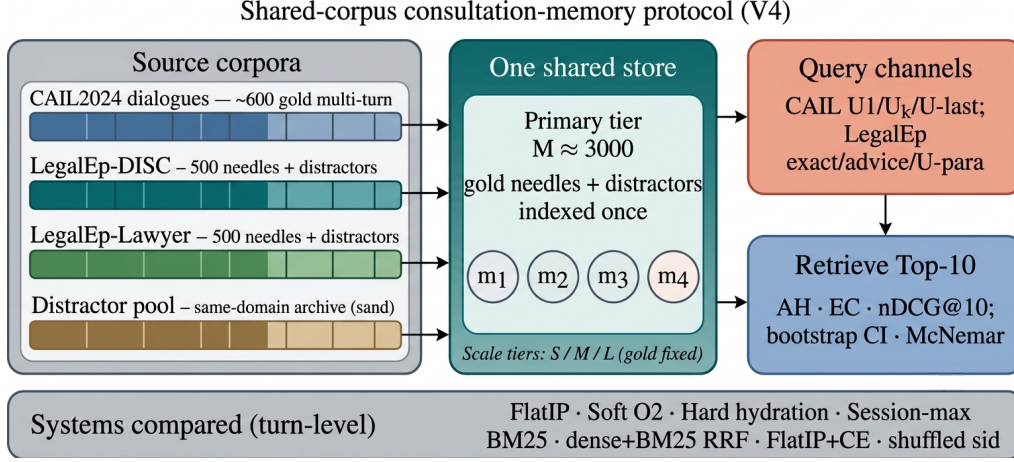


Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

230 4. Corpora and Evaluation Protocol

231 Figure 2 sketches the shared-corpus design used throughout the legal eval-
 232 uation. Gold needles and same-domain distractors occupy one store. Each
 233 query probes that shared archive under fixed interference conditions. Pri-
 234 vately rebuilt haystacks are avoided so that system comparisons isolate the
 235 retrieval operator.

236 4.1. Datasets and query channels

237 Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues.
 238 Gold needles are authentic preliminary and final dialogues, with conversation
 239 turns completed by label turns where applicable. Query channels cover the
 240 first user turn ($U1$), a later in-dialogue user turn ($U_k\text{-followup}$), the last user
 241 turn ($U\text{-last}$), and constrained paraphrases ($U\text{-para}$). Evidence defaults to
 242 assistant turns in the target session.

243 LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and
 244 Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is
 245 a natural (question, answer) pair with session identifier *sid*. Reconstruction
 246 applies length bounds and legal-content filters, excludes exam prompts, re-
 247 moves near-duplicates, separates needles from distractors under a fixed seed,
 248 and retains an audit subsample. After filtering, each LegalEp corpus retains

249 3500 passed sessions (500 needles, 2500 distractors at the $M=3000$ primary
250 tier).

251 *Query-channel policy (addressing exact-match shortcuts)*.. Exact replay uses
252 the stored user question and is reported only as a *diagnostic* upper bound
253 on surface-form recovery. Primary claims use paraphrase, Uk-followup /U-
254 last, and advice-recall queries whose surface form is not identical to the
255 stored question text; paraphrase and advice-recall templates are generated
256 under a fixed seed and audited on a subsample. LegalMem-MT reuses CAIL
257 multi-turn gold under the same store for same-session Soft O2 vs Soft O2-C
258 contrasts and for Mix reconstructions.

259 Archive size is varied with gold needles held fixed. Unless stated otherwise,
260 Soft O2 transfer tables use the $M \approx 3000$ tier; O1–O2 ablations use $M=400$,
261 $S=300$.

262 4.2. Metrics and baselines

263 Primary metrics are Answer Hit@ k (AH), Episode Completeness@ k (EC),
264 and normalized Discounted Cumulative Gain@ k (nDCG) at $k=10$. AH is
265 binary: whether a gold answer turn appears in the top- k . EC is the mean
266 coverage of gold turns within the target session (formerly reported as “session
267 recall” in early drafts; it is *not* binary Session Hit). Session Hit@ k (whether
268 any turn from the target *sid* appears) is reported only for diagnosis and is
269 distinct from EC. nDCG uses graded turn-level relevance with answer turns
270 preferred. Primary contrasts use bootstrap 95% confidence intervals (CIs)
271 and McNemar mid- p tests on paired per-query hits.

272 Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies
273 soft sibling score inheritance on the episodic pathway. Soft O2-C applies
274 the analogous rule within BIRCH clusters. Hard parent hydration expands
275 a hit to all siblings by copying the trigger score. Lexical controls include
276 BM25 over turns and over joint question–answer documents, together with
277 dense+BM25 Reciprocal Rank Fusion (RRF). A cross-encoder (CE) control
278 reranks the top-30 FlatIP candidates with BAAI/bge-reranker-v2-m3.
279 Negative and sensitivity controls include shuffled *sid*, a β sweep, and IVFPQ
280 as a quantization reference.

281 4.3. Fair Mix protocol for Soft O2-C

282 On fully same-session LegalMem grids, Soft O2 already recovers gold
283 answers that share *sid* with the query, so Soft O2-C has little exclusive head-
284 room and may inject thematic confusers. We therefore reconstruct the *same*

LegalEp and LegalMem archives under a Mix protocol: 70% of gold episodes are split into query-side S_a and evidence-side S_b with distinct session identifiers, while 30% remain intact. Distractors are unchanged. All operators—FlatIP, Soft O2, Soft O2-C, and gated Hybrid—share unrestricted dense retrieval on this store; we report overall AH and stratum AH on `cross_session` vs `same_session` queries.

Glue as a solvability bound (not a retrieval cheat). After BIRCH consolidation on the Mix store, each cross-session S_a/S_b split pair is *glued* into one cluster identifier. This step uses gold split metadata available only in the evaluation protocol—it does *not* inject labels at query time and is *not* used in real unsupervised clustering deployments. Its sole purpose is to establish a **solvability bound**: if Soft O2-C cannot recover cross-session gold when the semantic store *must* contain both halves of a split pair in one cluster, then failure reflects mechanism limits rather than accidental cluster separation. Glue therefore tests whether cluster soft binding *can* complete episodes when the slow pathway has the requisite thematic co-membership; natural same-cluster rates without glue are lower and are reported only as diagnostics (Section 6.3). Soft O2’s session membership constraint is unchanged—glue affects only whether Soft O2-C has a cluster in which to bind, not whether Soft O2 receives privileged label access. LegalEp Mix uses advice-recall queries; LegalMem Mix uses mid-split Uk-followup queries. External LongMemEval /LoCoMo numbers from prior retrieval runs on this codebase (QA correctness 70.7% / 68.2% on released test splits) are cited only as long-horizon memory context Wu et al. (2024); Maharana et al. (2024), not as Soft O2-C evidence on legal corpora.

5. Experiments

Unless noted, the embedding model is `nomic-embed-text-v2-moe` Nussbaum et al. (2025), $k=10$, seed 42. We report primary contrasts in figures and keep full numeric grids in the appendix. Result files and scripts are released with the accompanying repository so that each table maps to a public JSON artifact.

5.1. Implementation operators and Soft O2 ablation

Table 2 reports the shared-store ablation on DISC-Law and Lawyer-LLaMA ($M=400$, $S=300$). IVFPQ baseline Answer Hit is 0.540 / 0.397. O1 alone

319 yields a small gain on DISC (0.567) and a near-tie on Lawyer (0.397), confirm-
 320 ing that exact indexing is a necessary implementation prerequisite but insuffi-
 321 cient alone. O1+Soft O2 raises Answer Hit to 0.780 / 0.680 (+0.240 / +0.283
 322 over baseline) and Episode Completeness to 0.888 / 0.840—the primary al-
 323 gorithmic gain. O3 is enabled for all configurations in this ladder as a con-
 324 struction prerequisite; it does not alter retrieval scores.

Table 2: O1+Soft O2 ablation on the shared store ($M=400$, $S=300$, $k=10$). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus.

Corpus	System	EC@10	AH@10	nDCG@10
DISC-Law	IVFPQ baseline	0.770	0.540	0.782
	+ O1 FlatIP	0.782	0.567	0.790
	+ O1+O2 Soft O2	0.888	0.780	0.889
Lawyer-LLaMA	IVFPQ baseline	0.698	0.397	0.733
	+ O1 FlatIP	0.698	0.397	0.733
	+ O1+O2 Soft O2	0.840	0.680	0.856

325 5.2. Soft O2 on session (CAIL and LegalEp)

326 Figure 3 summarizes AH@10 and EC@10 on authentic CAIL multi-turn
 327 channels at $M \approx 3000$ under O1 FlatIP. Soft O2 lifts Answer Hit over FlatIP
 328 on every channel (U1 0.788 vs 0.570**; Uk 0.698 vs 0.270**; U-last 0.762 vs
 329 0.245**). Shuffled *sid* removes the gain (U1 0.452; Uk 0.230; U-last 0.200),
 330 tying the improvement to correct episode membership (RQ1–RQ2). Hard
 331 expansion can raise AH slightly above Soft O2 (e.g., U1 0.802 vs 0.788) while
 332 lowering EC (U1 0.578 vs Soft O2 0.664). Soft O2 therefore behaves as a
 333 rank-preserving binder: siblings enter under attenuated scores, and episode
 334 coverage remains higher than under hard score copy. On CAIL, nDCG often
 335 falls relative to FlatIP even as AH and EC rise, reflecting an availability–
 336 ranking trade-off. Soft O2 still exceeds FlatIP+CE on every CAIL channel
 337 (Table A.5). Full rows appear in Table A.1.

338 Figure 4 reports LegalEp transfer. On LegalEp-DISC exact replay Soft O2
 339 and FlatIP stay near-tied (0.638 vs 0.640), where both turns are already
 340 recoverable for many queries—consistent with treating exact as diagnostic.
 341 On Lawyer exact Soft O2 improves AH to 0.752 from FlatIP 0.692 (McNe-
 342 mar mid- $p=0.024^*$). Paraphrase yields the clearest LegalEp gains: Soft O2

CAIL2024 multi-turn results ($M \approx 3000$, turn-level systems)

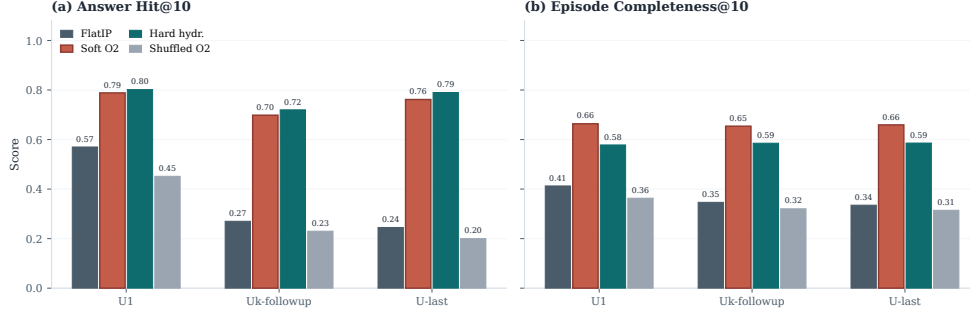


Figure 3: CAIL2024 multi-turn results at tier $M \approx 3000$. Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard expansion can maximize AH at a modest EC cost.

reaches 0.648 vs FlatIP 0.547 on DISC and 0.759 vs 0.568 on Lawyer, exceeding hard expansion on both corpora (mid- $p < 0.001^{**}$ on both corpora). Advice-recall queries omit the stored question surface form: Soft O2 raises AH to 0.428 from FlatIP 0.342 on DISC and to 0.472 from 0.332 on Lawyer, again above hard expansion (mid- $p < 0.001^{**}$), while shuffled *sid* falls toward or below FlatIP. Detailed grids are in Tables A.2–A.3.

5.3. Same-store cluster pathway validation

Table 3 contrasts Soft O2 and ungated Soft O2-C on standard same-session LegalEp /LegalMem stores (unrestricted dense; gold answers share *sid* with the query). Soft O2 dominates Soft O2-C on every reported channel (e.g., LegalMem U1 AH 0.903 vs 0.712; LegalEp-DISC exact 0.836 vs 0.744). The slow pathway is therefore *not* a free upgrade when episode membership already encodes the gold answer: cluster siblings inject thematic confusers that displace session-complete evidence.

Table 4 reports the complementary fair Mix setting (RQ3–RQ4): the same stores with 70% cross-session Sa/Sb gold and 30% intact same-session gold, unrestricted dense for every operator. Gated Hybrid Soft O2+Soft O2-C improves over Soft O2 on LegalEp-Lawyer overall (AH 0.534 vs 0.496; mid- $p = 0.010^{*}$) and on the cross-session stratum (0.523 vs 0.446; mid- $p < 10^{-4^{**}}$). On LegalEp-DISC, Hybrid is directionally above Soft O2 (0.486 vs 0.482) but not significant at $N = 500$. On LegalMem-MT, Hybrid attains the highest overall AH (0.646 vs Soft O2 0.618; mid- $p = 0.18$; cross-session mid- $p = 0.095$) while preserving most same-session hits (0.867 vs 0.887). Ungated Soft O2-C

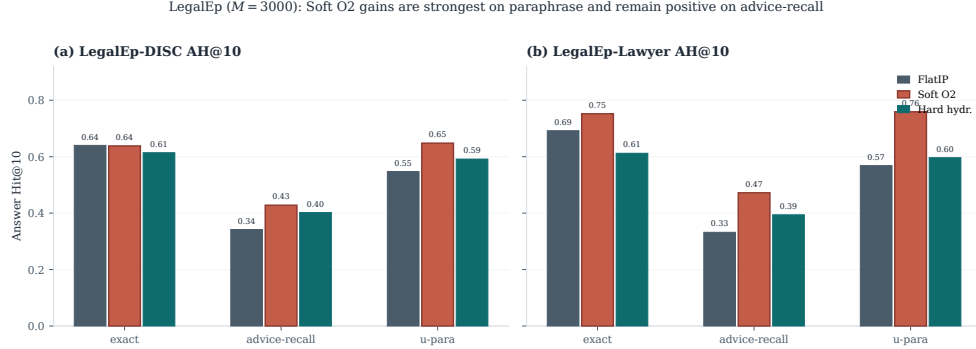


Figure 4: LegalEp AH@10 ($M=3000$, 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

Table 3: Same-session Soft O2 vs Soft O2-C on the shared LegalEp /LegalMem stores (AH@10). Soft O2-C is ungated cluster soft binding. Boldface: better AH within each row.

Corpus / channel	Soft O2	Soft O2-C
LegalEp-DISC / exact	0.836	0.744
LegalEp-DISC / u-para	0.807	0.730
LegalEp-DISC / advice	0.548	0.490
LegalEp-Lawyer / exact	0.818	0.710
LegalEp-Lawyer / u-para	0.817	0.733
LegalEp-Lawyer / advice	0.546	0.448
LegalMem-MT / U1	0.903	0.712
LegalMem-MT / Uk	0.793	0.385

can raise cross-session hits but often displaces same-session siblings; Hybrid triggers cluster binding only on direct dense hits. Together, Tables 3–4 show that the BIRCH pathway is effective precisely when gold evidence is not co-session with the query.

McNemar mid- p tests on paired Answer Hit; * $p<0.05$, ** $p<0.01$.

5.4. Controls, significance, scale, and QA audit

Lexical BM25 is strong on exact replay (DISC exact BM25-turn 0.802, BM25-joint 0.976) and weaker on advice-recall (0.382 / 0.432). Dense+BM25 RRF complements Soft O2 on surface-overlap channels. Cross-encoder reranking (bge-reranker-v2-m3, top-30) improves FlatIP inside the CE suite, yet

Table 4: Fair Mix protocol (same store, unrestricted dense; AH@10). Gold is a 70%/30% mix of cross-session Sa/Sb splits and intact same-session episodes. Soft O2-C denotes ungated cluster binding; Hybrid is Soft O2+gated Soft O2-C. Boldface: best AH within each corpus. Significance vs Soft O2: * $p<0.05$, ** $p<0.01$.

Corpus	System	AH	AH _{cross}	AH _{same}	mid- p vs Soft O2
LegalEp-DISC Mix	FlatIP	0.478	0.477	0.480	—
	Soft O2	0.482	0.443	0.573	—
	Soft O2-C	0.410	0.443	0.333	0.0004**
	Hybrid	0.486	0.469	0.527	0.8099
LegalEp-Lawyer Mix	FlatIP	0.462	0.454	0.480	—
	Soft O2	0.496	0.446	0.613	—
	Soft O2-C	0.464	0.491	0.400	0.0857
	Hybrid	0.534*	0.523	0.560	0.0105*
LegalMem-MT Mix	FlatIP	0.534	0.526	0.553	—
	Soft O2	0.618	0.503	0.887	—
	Soft O2-C	0.500	0.537	0.413	0.0000**
	Hybrid	0.646	0.551	0.867	0.1837

376 remains below Soft O2 on CAIL (Uk FlatIP+CE 0.375 vs Soft O2 0.698;
 377 Table A.5). A β sweep peaks near $\{0.9, 0.95, 0.98\}$ and weakens at $\beta=0.5$
 378 (Table A.6). McNemar mid- p tests (Table A.7) show Soft O2 versus FlatIP
 379 highly significant on all CAIL channels and on LegalEp paraphrase /advice-
 380 recall.

381 Figure 5 and Table A.8 report quality and warm-query latency on LegalEp-
 382 DISC exact ($Q=100$, three repeats; $M \in \{100, 400, 1600, 6400\}$). Soft O2
 383 retains an AH advantage at each tier; IVFPQ collapses at $M=1600$ (AH
 384 0.330), motivating O1; absolute latency grows with M . We treat O1 FlatIP
 385 as an engineering prerequisite for Soft O2 at these scales and avoid claiming
 386 deployment readiness beyond the measured curve.

387 *End-to-end QA audit (summary)*.. A dual-protocol QA audit ($N=270$ per
 388 corpus; generator **qwen3:14b**, independent judge **qwen3:32b**) links retrieval
 389 quality to downstream answerability; full protocol and stratified results ap-
 390 pear in Section 6.2. On an older revision-protocol store ($M=400$, five seeds),
 391 Soft O2 AH on LegalEp-DISC is 0.851 ± 0.008 versus FlatIP 0.698 ± 0.040 .

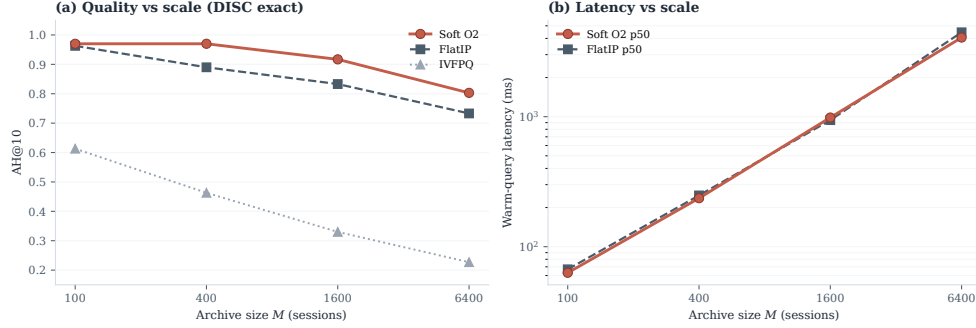


Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as M grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate M .

6. Discussion

6.1. Findings and scope

RQ1–RQ2 confirm that episode incompleteness dominates under same-domain interference. O1 restores score fidelity as an implementation prerequisite; Soft O2 supplies the primary algorithmic gain through soft sibling inheritance; O3 enables archive construction at scale. On CAIL, Soft O2 raises Answer Hit by large margins (U1 +0.218, Uk +0.428, U-last +0.517) with collapsing shuffled-*sid* controls, and keeps higher EC than hard expansion despite occasional AH ties. LegalEp paraphrase and advice-recall—not exact replay—carry the primary Soft O2 transfer claims. RQ3 shows that the BIRCH slow pathway benefits Soft O2-C only when gold is not already co-session; Mix Hybrid Soft O2+Soft O2-C recovers that regime without seed-forcing Soft O2 to zero. RQ4: Soft O2 (session) and Soft O2-C /Hybrid (cluster) are complementary operators within one BIMS-LEGAL dual-store. Sparse, fusion, and cross-encoder baselines situate Soft O2 among standard IR alternatives Robertson and Zaragoza (2009); Cormack et al. (2009); Nogueira et al. (2019).

6.2. RAG application perspective

Consultation-memory retrieval is judged downstream by whether an LLM can answer from recovered evidence rather than hallucinate. We audit this end-to-end with a dual-protocol pipeline: generator `qwen3:14b` and independent judge `qwen3:32b` are distinct models. Per corpus we pool $N=270$

judged instances as P1 $n=120$ plus P2 $n=150$ (folder `legal_qa_n270`). Table 5 summarizes pooled Evidence-Answerability Rate (EAR) and stratified independent-judge scores. When Soft O2 retrieves complete episodes—both question and answer turns in top- k —the generator receives grounded prior advice and judged answerability rises (pooled EAR 0.867 on DISC and 0.859 on Lawyer; Wilson 95% CI [0.83, 0.90] / [0.82, 0.89]). Follow-up queries remain the hardest stratum (independent-judge EAR 0.517 / 0.480), consistent with episodic incompleteness persisting even when surface-form overlap is low.

A blind two-annotator legal audit ($n=60$ /corpus) yields Cohen’s $\kappa=0.71$ / 0.68 and separates retrieval from generation failures (retrieval-failure share 0.283 / 0.317). Wrong-session contamination rates stay low (0.083 / 0.100), suggesting that complete episode retrieval under Soft O2 reduces a common RAG failure mode in which the model answers from a neighbouring client’s matter. Hallucinated-legal-basis rates (0.117 / 0.133) remain non-zero, indicating that grounding in retrieved episodes does not guarantee statutory correctness—a scope boundary for deployment. AH and EAR measure evidence availability and judged answerability respectively; statutory correctness, citation validity, and professional liability lie outside this evaluation.

Table 5: End-to-end QA audit ($N=270$ per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`). EAR: Evidence-Answerability Rate (judge score ≥ 0.7). Stratified judge scores: $n=90$ /corpus across exact, paraphrase, and follow-up protocols.

Corpus	Pooled EAR	Stratified judge	Human κ
DISC-Law	0.867	0.757	0.71
Lawyer-LLaMA	0.859	0.729	0.68

6.3. Limitations

LegalEp episodes are consultation pairs; multi-turn authenticity remains reserved for CAIL. Mix glue establishes a solvability bound for Soft O2-C evaluation (Section 4.3); natural same-cluster rates without glue are lower and are diagnostics only—real unsupervised BIRCH clustering would not receive split-pair metadata at build time. Soft O2 depends on correct session identifiers; Soft O2-C depends on cluster quality. LLM paraphrases need human audit. Statutory correctness, citation validity, and professional liability remain unevaluated. Multiple channels inflate comparison multiplicity even with McNemar tests.

443 *Latency and deployment scope..* Scale curves cover $M \in$
444 $\{100, 400, 1600, 6400\}$ under a fixed hardware stack (Table A.8). At
445 $M=6400$, warm-query p50 latency exceeds 4 s (4061 ms for Soft O2; 4458 ms
446 for FlatIP)—unacceptable for millisecond interactive search. BIMS-LEGAL
447 should therefore be positioned as **high-precision offline or quasi-**
448 **real-time consultation memory recovery**—e.g., pre-meeting case
449 review, audit replay, or batch RAG over a firm’s advice archive—not as
450 a sub-millisecond search engine. Future work will combine approximate
451 nearest-neighbour indexes (e.g., HNSW) with Soft O2 and Soft O2-C
452 binding layered on top of faithful candidate retrieval, preserving episode
453 completion while reducing query latency. Behavior at roughly 10^5 sessions is
454 not measured, and we do not claim asymptotic scalability from O3 wall-clock
455 alone.

456 7. Conclusion

457 BIMS-LEGAL targets consultation-memory retrieval under same-domain
458 interference in Chinese legal archives through a dual-store design: an episodic
459 turn index paired with a BIRCH semantic store. Soft O2 session soft binding
460 and Soft O2-C with gated Hybrid cluster binding are the algorithmic core;
461 O1 exact indexing and O3 bulk-load consolidation are implementation oper-
462 ators that enable the shared-store ablations reported here. Under a shared-
463 corpus protocol spanning CAIL multi-turn dialogues and rebuilt LegalEp
464 episode corpora, Soft O2 improves Answer Hit over FlatIP on multi-turn,
465 paraphrase, and advice-recall channels with collapsing shuffled-*sid* controls,
466 and yields higher episode completeness than hard expansion on CAIL. Same-
467 store Soft O2-C underperforms Soft O2 when gold shares *sid*; under fair
468 Mix reconstructions, gated Hybrid Soft O2+Soft O2-C recovers cross-session
469 gold while Soft O2 remains strong on intact same-session gold. End-to-end
470 QA audit links complete episode retrieval to judged answerability and low
471 wrong-session contamination in downstream generation. Sparse, fusion, and
472 cross-encoder baselines situate Soft O2 among standard IR alternatives.

473 Declarations

474 *Funding*

475 Computational resources were provided by the Data Law Laboratory,
476 China University of Political Science and Law.

477 *Conflict of interest*

478 The authors declare no competing interests.

479 *Ethics approval*

480 This study uses publicly released research corpora and excludes confidential client data. The system is a research prototype for consultation-memory retrieval.

483 *Data availability*

484 Code, processed evaluation corpora, and primary result files are publicly available at <https://github.com/Kilimajaro/soft-episode-binding-legal-memory>. Upstream legal corpora: CAIL2024 consultation tracks; Hugging Face ShengbinYue/DISC-Law-SFT and Skepsun/lawyer_llama_data.

489 *Code availability*

490 The BIMS-LEGAL implementation (O1 FlatIP, Soft O2, O3 bulk-load, Soft O2-C), LegalMem/LegalEp pipelines, Mix builders, and table/figure scripts are released in the same repository with JSON artifacts for each main table.

494 *Author contributions*

495 L.X. (University of Winnipeg) designed BIMS-LEGAL and the Soft O2 /Soft O2-C evaluation, implemented the system, ran the experiments, and wrote the manuscript. L.H. (corresponding author; CUPL Data Law Lab and Institute for Data Law) provided methodology guidance, revision, and supervision.

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609 Appendix A. Numeric grids and controls

610 Main Soft O2 claims are visualized in Figures 3–5; O1–O2 and Mix
611 Soft O2-C tables appear in the main text. Tables below retain full Soft O2
612 numeric rows for reproducibility.

Table A.1: CAIL2024 shared-corpus results ($M \approx 3000$). Boldface marks the best AH@10 within each channel (ties bolded jointly). **: Soft O2 vs FlatIP, McNemar mid- $p < 0.01$. Hard hydration coincides with session-max aggregation on these channels; only hard hydration is shown.

Channel	System	AH@10	EC@10	nDCG@10	95% CI (AH)
U1	FlatIP	0.570	0.413	0.805	[0.530,0.610]
	Soft O2	0.788**	0.664	0.772	[0.757,0.820]
	Hard hydr.	0.802	0.578	0.840	[0.768,0.835]
	Shuffled O2	0.452	0.363	0.664	[0.410,0.492]
Uk-followup	FlatIP	0.270	0.347	0.884	[0.237,0.307]
	Soft O2	0.698**	0.654	0.786	[0.660,0.735]
	Hard hydr.	0.720	0.585	0.841	[0.685,0.753]
	Shuffled O2	0.230	0.320	0.718	[0.198,0.262]
U-last	FlatIP	0.245	0.335	0.891	[0.212,0.280]
	Soft O2	0.762**	0.659	0.779	[0.725,0.797]
	Hard hydr.	0.790	0.586	0.844	[0.757,0.822]
	Shuffled O2	0.200	0.315	0.726	[0.172,0.232]

Table A.2: LegalEp-DISC ($M=3000$, 500 needles). Boldface: best AH@10 within each channel. * $p<0.05$, ** $p<0.01$ (Soft O2 vs FlatIP, McNemar mid- p). Hard hydration coincides with session-max aggregation; only hard hydration is shown.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.640	0.741	—
	Soft O2	0.638	0.780	—
	Hard hydr.	0.614	0.598	—
	Shuffled O2	0.448	0.645	—
advice-recall	FlatIP	0.342	0.438	0.506
	Soft O2	0.428**	0.503	0.419
	Hard hydr.	0.402	0.405	0.474
	Shuffled O2	0.310	0.421	0.402
u-param	FlatIP	0.547	0.681	0.705
	Soft O2	0.648**	0.735	0.583
	Hard hydr.	0.592	0.571	0.651
	Shuffled O2	0.469	0.610	0.558

Table A.3: LegalEp-Lawyer ($M=3000$, 500 needles). Boldface: best AH@10 within each channel. * $p<0.05$, ** $p<0.01$ (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.692	0.806	0.844
	Soft O2	0.752*	0.848	0.687
	Hard hydr.	0.612	0.615	0.754
	Shuffled O2	0.570	0.747	0.653
advice-recall	FlatIP	0.332	0.464	0.589
	Soft O2	0.472**	0.540	0.460
	Hard hydr.	0.394	0.403	0.488
	Shuffled O2	0.348	0.467	0.436
u-param	FlatIP	0.568	0.718	0.834
	Soft O2	0.759**	0.819	0.645
	Hard hydr.	0.596	0.594	0.707
	Shuffled O2	0.592	0.712	0.627

Table A.4: BM25 turn/joint and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported cells.

Corpus	Channel	BM25-turn	BM25-joint
LegalEp-DISC	exact	0.802	0.976
LegalEp-DISC	advice-recall	0.382	0.432
LegalEp-DISC	exact (dense RRF)	0.780	—
LegalEp-DISC	advice (dense RRF)	0.394	—
CAIL	U1 / Uk / U-last	0.748 / 0.482 / 0.413	0.993 / 0.840 / 0.823
LegalEp-Lawyer	exact / advice	0.784 / 0.460	0.998 / 0.652

Table A.5: Cross-encoder control (AH@10). FlatIP and FlatIP+CE are paired in the CE suite (**bge-reranker-v2-m3** over top-30 dense candidates). Boldface: best AH within each row. Soft O2 reproduces the primary evaluation; FlatIP here may differ slightly from primary FlatIP due to independent CE-suite re-indexing.

Corpus / channel	FlatIP	FlatIP+CE	Soft O2
CAIL / U1	0.585	0.688	0.788
CAIL / Uk	0.265	0.375	0.698
CAIL / U-last	0.228	0.342	0.762
LegalEp-DISC / exact	0.572	0.722	0.638
LegalEp-Lawyer / exact	0.636	0.708	0.752

Table A.6: Soft O2 AH@10 versus β on the primary shared-corpus evaluation. Peak performance lies in $\{0.9, 0.95, 0.98\}$; aggressive attenuation ($\beta=0.5$) weakens binding.

Corpus	0.5	0.7	0.9	0.95	0.98	1.0
CAIL / Uk	0.323	0.537	0.697	0.700	0.698	0.662
LegalEp-Lawyer / exact	0.608	0.654	0.754	0.754	0.752	0.752
LegalEp-Lawyer / u-para	0.631	0.637	0.717	0.745	0.759	0.749
LegalEp-DISC / u-para	0.551	0.551	0.618	0.648	0.648	0.642

Table A.7: McNemar mid- p on paired Answer Hit (Soft O2 vs FlatIP / hard hydration / FlatIP+CE).

Corpus / channel	Contrast	ΔAH	mid- p
CAIL / Uk	O2 vs FlatIP	+0.428	8.99e-48
CAIL / Uk	O2 vs Hard hydr.	-0.022	0.243
CAIL / U1	O2 vs FlatIP	+0.218	5.19e-18
CAIL / U1	O2 vs Hard hydr.	-0.013	0.435
CAIL / U-last	O2 vs FlatIP	+0.517	5.51e-65
CAIL / U-last	O2 vs Hard hydr.	-0.028	0.0982
CAIL / Uk	O2 vs FlatIP+CE	+0.323	2.50e-26
CAIL / U1	O2 vs FlatIP+CE	+0.100	3.51e-05
CAIL / U-last	O2 vs FlatIP+CE	+0.420	3.92e-41
LegalEp-DISC / exact	O2 vs FlatIP	-0.002	0.934
LegalEp-DISC / exact	O2 vs Hard hydr.	+0.024	0.302
LegalEp-Lawyer / exact	O2 vs FlatIP	+0.060	0.0239
LegalEp-Lawyer / exact	O2 vs Hard hydr.	+0.140	3.50e-10
LegalEp-DISC / u-para	O2 vs FlatIP	+0.101	7.95e-05
LegalEp-DISC / u-para	O2 vs Hard hydr.	+0.056	0.035
LegalEp-Lawyer / u-para	O2 vs FlatIP	+0.191	2.10e-15
LegalEp-Lawyer / u-para	O2 vs Hard hydr.	+0.163	7.57e-09
LegalEp-DISC / advice	O2 vs FlatIP	+0.086	1.25e-04
LegalEp-DISC / advice	O2 vs Hard hydr.	+0.026	0.215
LegalEp-Lawyer / advice	O2 vs FlatIP	+0.140	1.18e-10
LegalEp-Lawyer / advice	O2 vs Hard hydr.	+0.078	5.72e-04

Table A.8: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each M block; lowest p50/p95 within each M block.

M	Mode	AH@10	Build (s)	p50 (ms)	p95 (ms)
100	FlatIP	0.963	3.0	67	80
100	IVFPQ	0.613	25.2	40	54
100	Soft O2	0.970	3.2	63	69
400	FlatIP	0.890	26.5	248	305
400	IVFPQ	0.463	89.5	171	204
400	Soft O2	0.970	30.9	236	313
1600	FlatIP	0.833	359.8	941	1278
1600	IVFPQ	0.330	251.4	669	751
1600	Soft O2	0.917	393.7	985	1172
6400	FlatIP	0.733	4722.8	4458	4969
6400	IVFPQ	0.227	1072.1	2182	3021
6400	Soft O2	0.803	5245.7	4061	4503